6. 设出数 m, [illegible] 每只卵数 n_i

即 $m \sim P(\lambda)$, $n_i \sim P(\mu)$

$\therefore E(mn) = \lambda \cdot \mu$.

$\therefore P(\sum_{i=1}^{m} n_i = k) = \sum_{i=0}^{\infty} \frac{i^i e^{-\lambda}}{i!} \cdot \frac{(i\mu)^k \cdot e^{-i\mu}}{k!} \quad e^{-(\lambda\mu)} \cdot \frac{(\lambda\mu)^k}{k!}$

$= e^{-\lambda} \cdot \frac{\mu^k}{k!} \cdot \sum_{i=0}^{\infty} \frac{\lambda^i \cdot i^k}{i!}$

[illegible]

9.

(1) $\because N_i \sim P(\lambda_i)$

$\therefore E(Z_j) = \sum_{i=1}^{j-1} E(N_i) p_{ij}$

$= \sum_{i=1}^{j-1} \lambda_i \cdot p_{ij}$

(2) [illegible] $\because$ 二次随机

$\therefore Z_j$ 满足参数为 $\sum_{i=1}^{j-1} \lambda_i \cdot p_{ij}$ 的泊松分布.

(3) $\because Z_k$ 满足参数为 $\sum_{i=1}^{k-1} \lambda_i \cdot p_{ik}$ 的泊松分布.

$\therefore P(Z_j = m, Z_k = n) = P(Z_j = m) \cdot P(Z_k = n)$

$= e^{-\lambda_1'} \cdot \frac{\lambda_1'^m}{m!} \cdot e^{-\lambda_2'} \cdot \frac{\lambda_2'^n}{n!}$

$\lambda_1' = \sum_{i=1}^{j-1} \lambda_i \cdot p_{ij} \qquad \lambda_2' = \sum_{i=1}^{k-1} \lambda_i \cdot p_{ik}$

WEIYANG COLLEGE TSINGHUA
清华大学未央书

作业纸

班级： 姓名： 学号： 科目： 第 页

10. $\because N \sim P(\lambda)$

$\therefore E(X) = p\lambda$

$\therefore P(X=k) = e^{-p\lambda} \cdot \frac{(p\lambda)^k}{k!}$

$E(Y) = \lambda \cdot (1-p)$

$\therefore P(Y=k) = e^{-\lambda(1-p)} \frac{(\lambda(1-p))^k}{k!}$

(2) $\because P(X=k_1, Y=k_2) = p^{k_1} \cdot (1-p)^{k_2} \cdot e^{-\lambda} \cdot \frac{\lambda^{k_1+k_2}}{(k_1+k_2)!} \cdot C_{k_1+k_2}^{k_1}$

而 $P(X=k_1) \cdot P(Y=k_2) = e^{-(\lambda)} \cdot \frac{\lambda^{k_1+k_2} \cdot p^{k_1}(1-p)^{k_2}}{k_1! \cdot k_2!}$

$\therefore$ 独立.

12. $\because M = \max(X_1, X_2)$, $N = \min(X_1, X_2)$

$\therefore P(X_1=m, X_2=n) = \frac{\lambda^{m+n} \cdot e^{-(m+n)}}{m!n!}$ (不妨设 $m \geq n$),

$\therefore P(M=m) = 2 \cdot \frac{\lambda^m e^{-m}}{m!} \cdot \sum_{i=0}^{m-1} \frac{\lambda^i e^{-\lambda}}{i!} + \frac{\lambda^{2m} \cdot e^{-2m}}{(m!)^2}$

$P(N=n) = 2 \cdot \frac{\lambda^n e^{-n}}{n!} \cdot \sum_{i=n+1}^{\infty} \frac{\lambda^i e^{-i}}{i!} + \frac{\lambda^{2n} \cdot e^{-2n}}{(n!)^2}$

WEIYANG COLLEGE TSINGHUA UNIVERSITY
清华大学未央书院

作业纸

班级： 姓名： 学号： 科目： 第 页

13. $\mathrm{Cov}(N_t, N_s) = E(N_t N_s) - E(N_t)E(N_s)$

$\because E(N_t) = \lambda t, \quad E(N_s) = \lambda \cdot s$

且 $E(N_t N_s) = E(N(t)\cdot(N(t)+N(t,s))]$

$= E(N^2(t)) + E(N(t))\cdot(s-t)\cdot\lambda$

$\therefore \mathrm{Cov}(N_t, N_b) = \mathrm{Var}(N(t)) = \lambda.$

14. $\therefore \lambda \cdot t = 0.1 \cdot 60 = 6$

$\therefore P(A) = P(N<3)$

$= \sum_{i=0}^{2} e^{-6}\cdot\frac{6^i}{i!}$

$= e^{-6}\cdot 25$

$\approx 0.062.$

15. $\because E(\dot{n}) = 60\cdot 3\cdot 60 = 10800$

$\therefore E(\eta_1) = 1.5$

$\therefore E(S) = 16200$

又$\because$ 相互独立, $\therefore D(S) = 16200$

$\therefore \sigma = 127.3$

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清华大学未央书

作业纸

班级： 姓名： 学号： 科目： 第 页

1b.

$$\because P(Y_t=k)=\sum_{n=0}^{\infty} P(N_t=n)\cdot P(Y_t=k\mid N_t=n)$$

$$=\sum_{n=0}^{\infty}\frac{(\lambda t)^n e^{-\lambda t}}{n!}\cdot\frac{n!\,p^k(1-p)^{n-k}}{k!(n-k)!}$$

$$=\frac{(\lambda tp)^k}{k!}e^{-\lambda tp},\quad \text{即为泊松过程.}$$

$$\therefore E(Y_t)=\lambda pt,\qquad D(Y_t)=\lambda pt$$

$$P(Y_t=0)=e^{-\lambda pt}$$

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